Team Sienna's

Final Solution

Transforming VADOC Procurement Through Data & Decision Intelligence

Meet Our Team: Team Sienna

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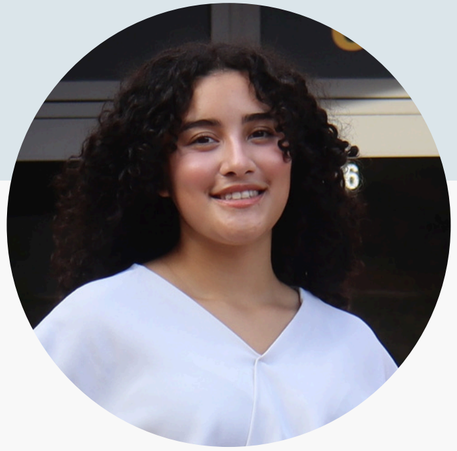


Tatiana White

Growth

The Problem We Addressed

Limited visibility into:

- Vendor performance
- Spending patterns
- Order activity

Lack of tools for:

- SWAM compliance tracking (42% target)
- Future cost forecasting

Decision-making is often:

- Manual
- Time-consuming
- Inconsistent

Our Objective:

- Build a centralized BI solution
- Improve transparency, efficiency, and decision-making
- Support data-driven procurement strategies

Who We Designed This For

Analyst

- Monitor procurement activity and order status
- Track vendor usage and approval timelines
- Stay organized and informed for daily operations

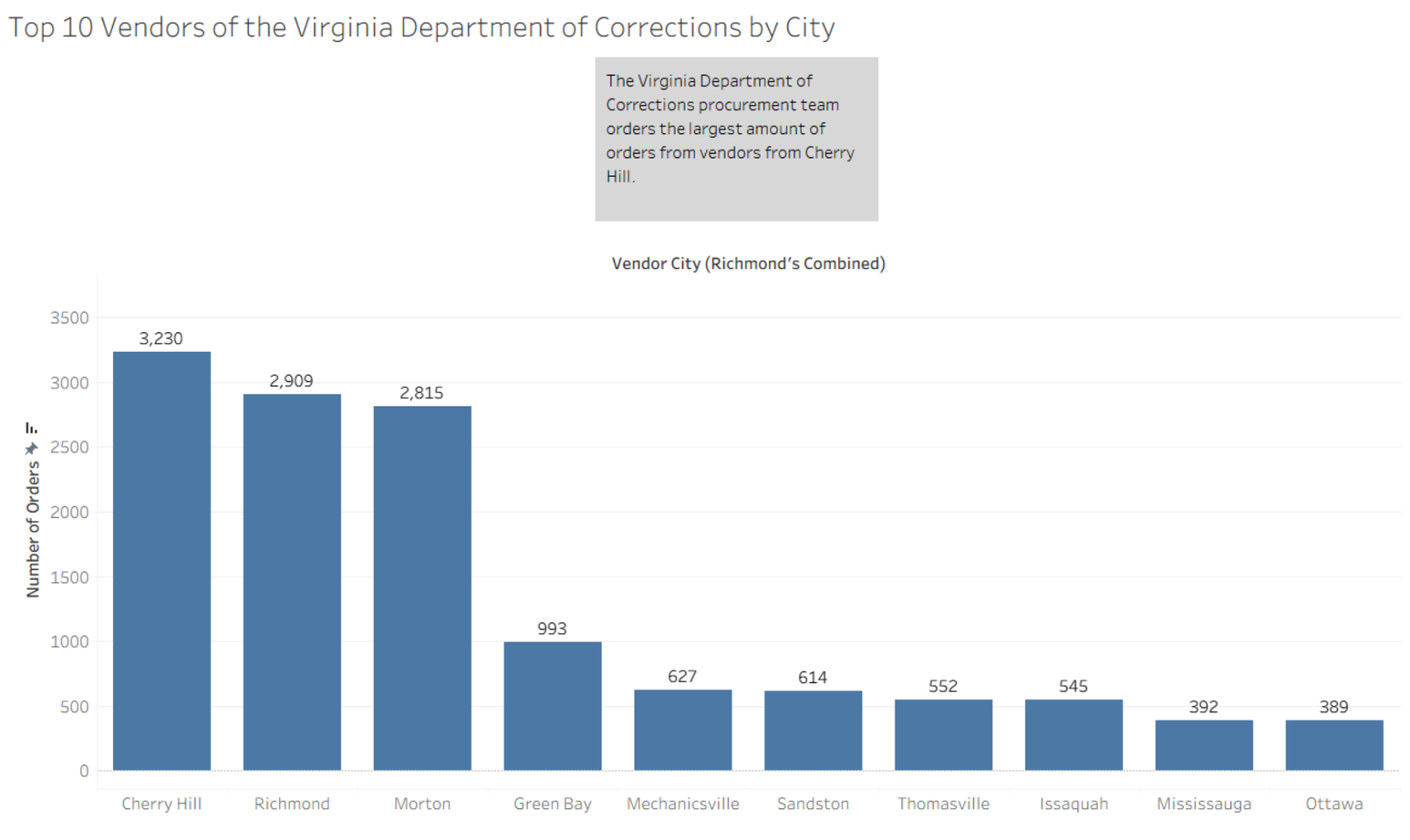
Leader

- Identify vendor trends across facilities
- Evaluate vendor dependency and spending patterns
- Make strategic sourcing decisions

Decision-maker

- Compare planned vs actual spending
- Validate procurement decisions with data
- Ensure cost-effectiveness and accountability

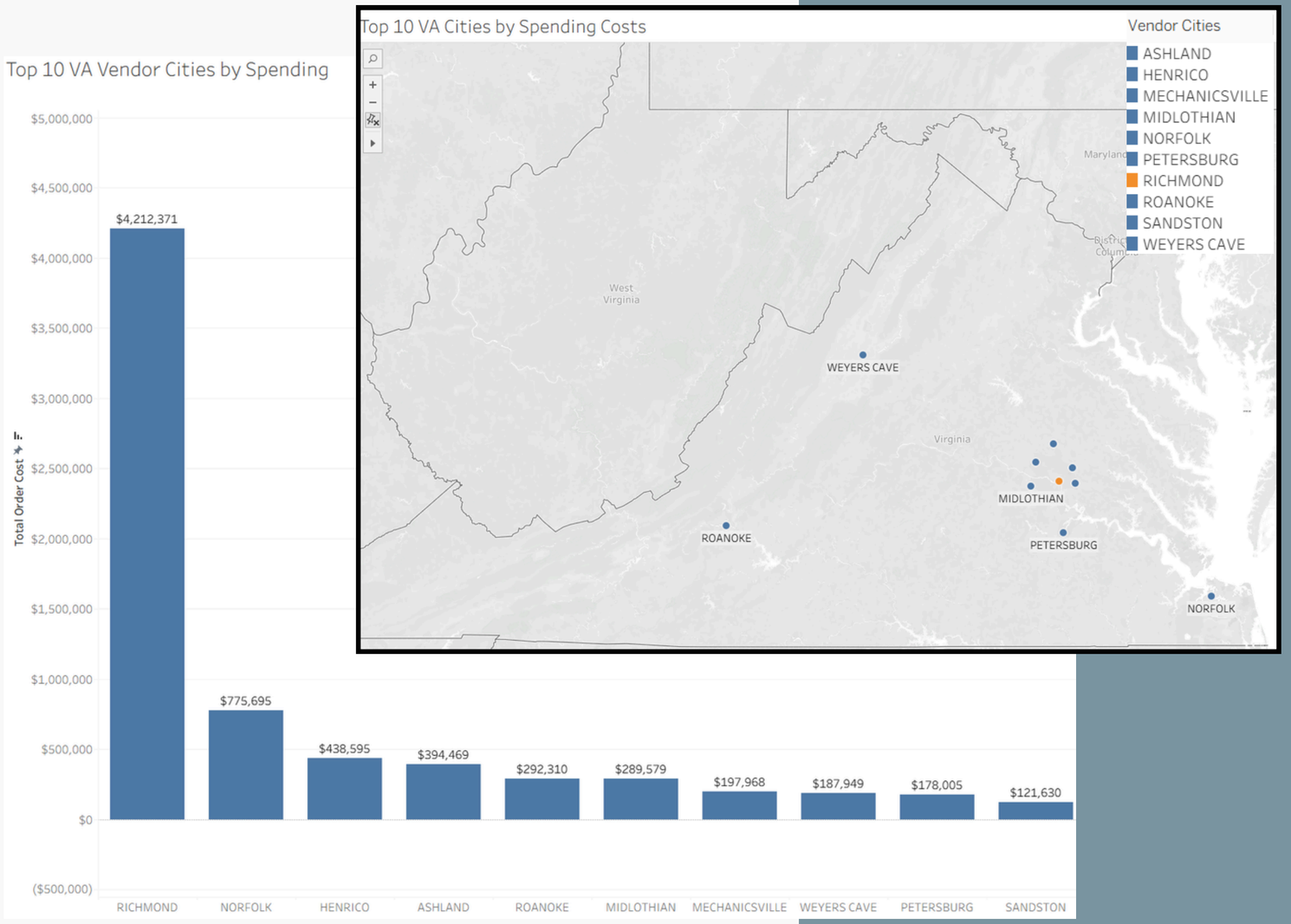
Improving Procurement Visibility



- **Purpose**:
  - Establish a baseline view of vendor activity concentration
- **What this reveals:**
  - Procurement activity is not evenly distributed across cities
  - High-volume cities indicate core supply hubs
- **Key Insight:**
  - Concentration in a few cities creates operational dependency risk
- **Business Value (Analyst):**
  - Enables faster identification of supply bottlenecks
  - Supports better coordination of incoming orders
  - Builds foundation for deeper spending + vendor analysis

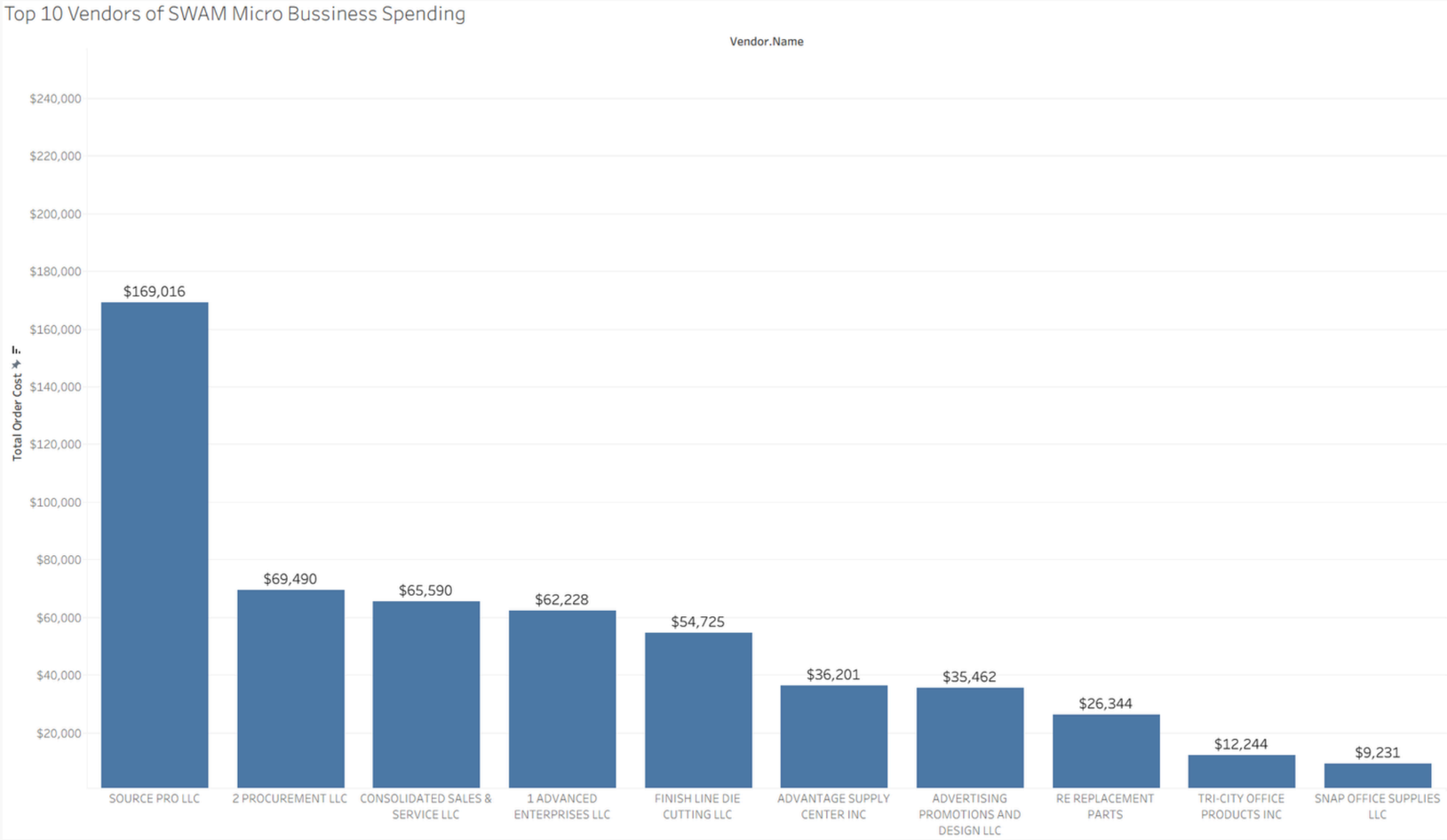
Understanding Spending Patterns

- **Purpose**:
  - Move from activity tracking → financial impact analysis
- **What this reveals:**
  - Spending is heavily concentrated in a few geographic areas
  - Proximity to HQ influences vendor selection behavior
- **Key Insight:**
  - High reliance on nearby vendors may limit cost efficiency and competition
- **Business Value (Leader):**
  - Supports strategic sourcing decisions
  - Highlights opportunities to:
  - Reduce over-reliance on single regions
  - Explore alternative vendors for cost optimization



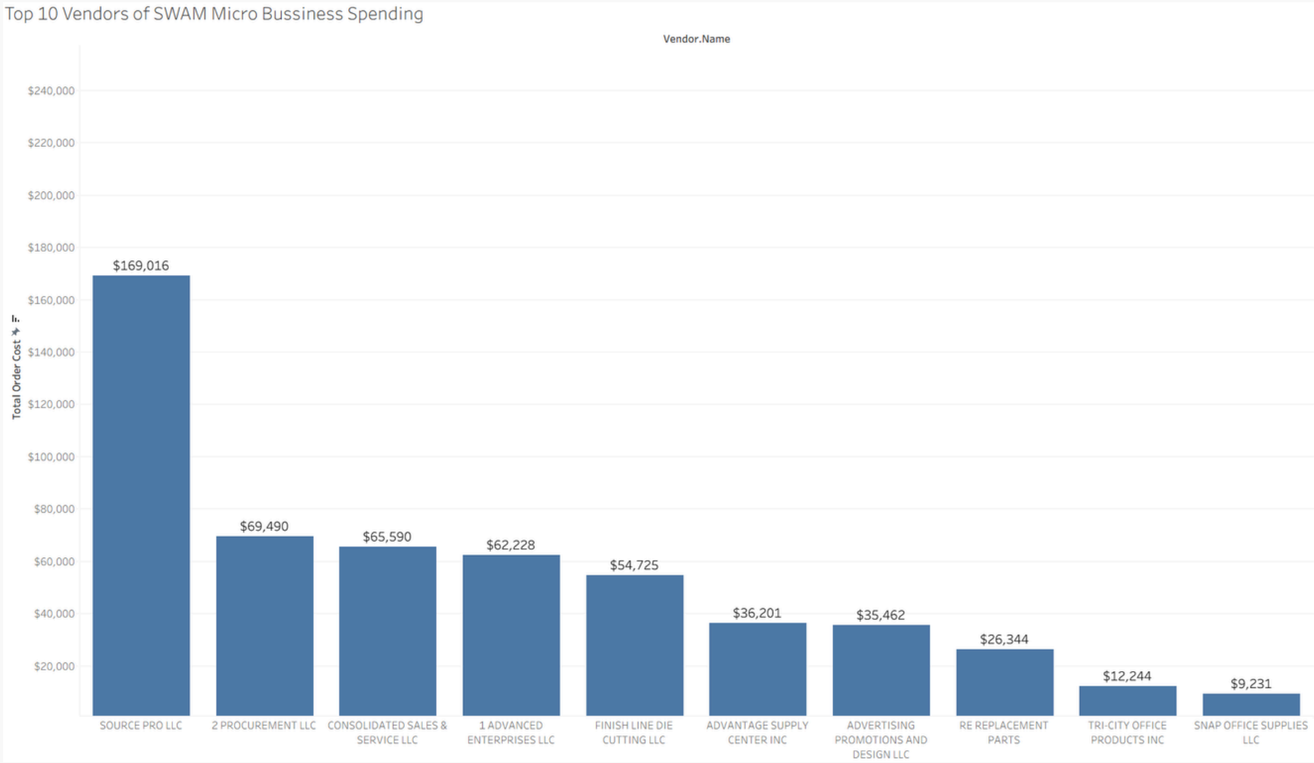
SWAM Vendor Analysis

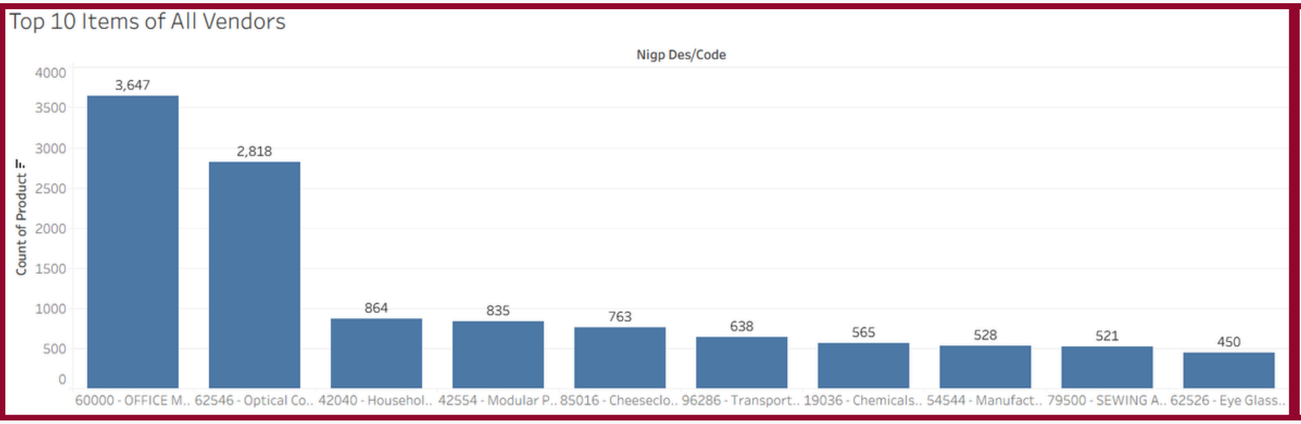
- **Purpose**:
  - Evaluate how procurement dollars are distributed within SWAM vendors
- **What this reveals:**
  - Spending is highly skewed toward a small number of vendors
  - Presence of outliers signals uneven vendor utilization
- **Key Insight:**
  - Over-concentration reduces supplier diversity and resilience
- **Business Value (Decision-maker):**
  - Validates whether spending aligns with equity + compliance goals
  - Identifies need to:
  - Expand participation across more SWAM vendors
  - Reduce dependency on top suppliers

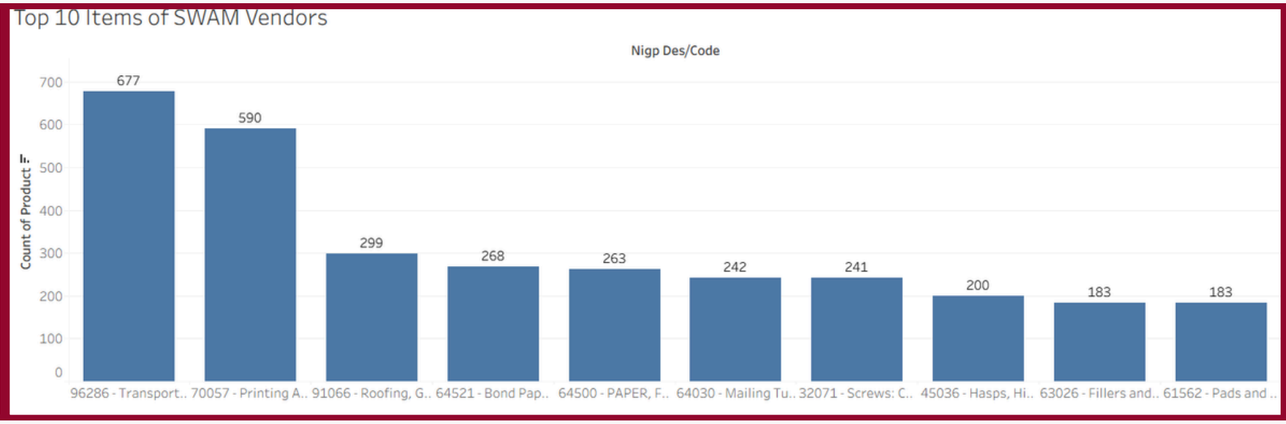


SWAM vs Non-SWAM Analysis

- **Purpose**:
  - Compare actual purchasing behavior vs compliance goals
- **What this reveals:**
  - Many high-demand categories are still fulfilled by non-SWAM vendors
  - SWAM participation is not aligned with spending patterns
- **Key Insight:**
  - There is a clear gap between current spending and the 42% SWAM target
- **Business Value (Leader + Decision Maker):**
  - Identifies where spending can realistically be shifted
  - Supports recommendations to:
  - Reallocate purchases in specific categories
  - Increase SWAM vendor inclusion without disrupting operations

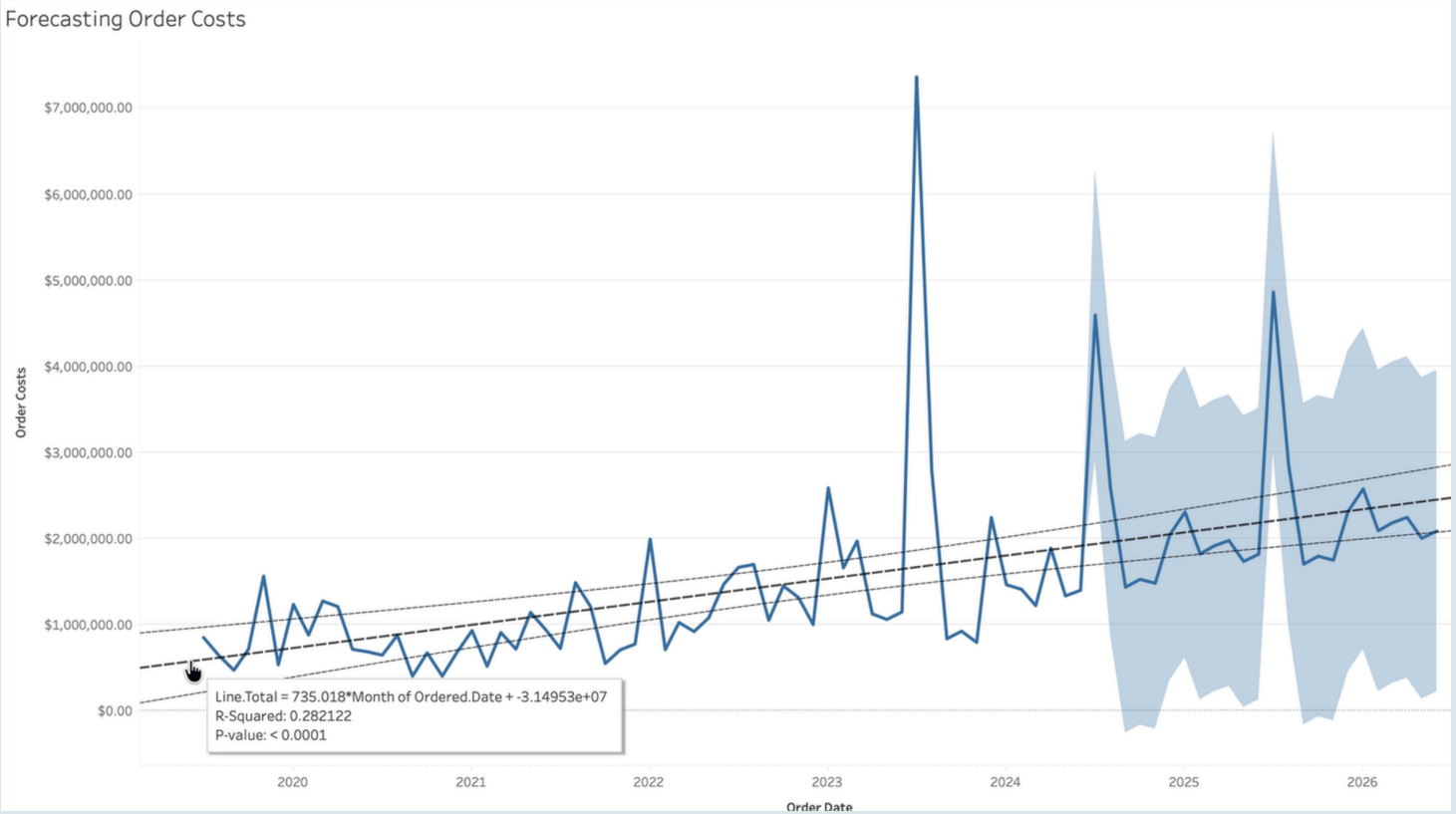






Forecasting Procurement Costs

- **Purpose:**
  - Extend analysis → future cost planning
- **What this reveals:**
  - Costs show seasonal patterns tied to fiscal cycles
  - July (Start of Fiscal Year) spikes align with budget resets
- **Key Insight:**
  - Costs are trending upward with high variability
- **Model Notes:**
  - Uses Tableau time-series forecasting (exponential smoothing)
  - Shaded area = confidence interval (uncertainty)
  - Detects trend + recurring seasonal spending patterns
  - Best used for planning directionally, not exact dollar predictions
- **Business Value (Decision-maker):**
  - Supports budget planning
  - Anticipates spending spikes + volatility
  - Enables proactive decision-making



Actionable Recommendations

1. Increase SWAM Vendor Participation:

- Shift spending in high-value categories
- Target underrepresented areas

2. Reduce Vendor Dependency

- Diversify vendor base across regions
- Improve procurement flexibility

3. Leverage Predictive Analytics

- Use forecasts for budget planning
- Anticipate seasonal spending spikes

4. Improve Data Quality & Standardization

- Address missing or inconsistent data
- Standardize data preparation processes

Why Our Team & Solution

Our team applied a disciplined Agile Scrum approach to iteratively transform raw procurement data into reliable, insight-driven dashboards using Excel for data preparation and Tableau for visualization and forecasting. By focusing on clarity, usability, and decision relevance, our solution goes beyond reporting to deliver actionable insights that users can trust, adopt, and use to improve procurement outcomes.

Thank you